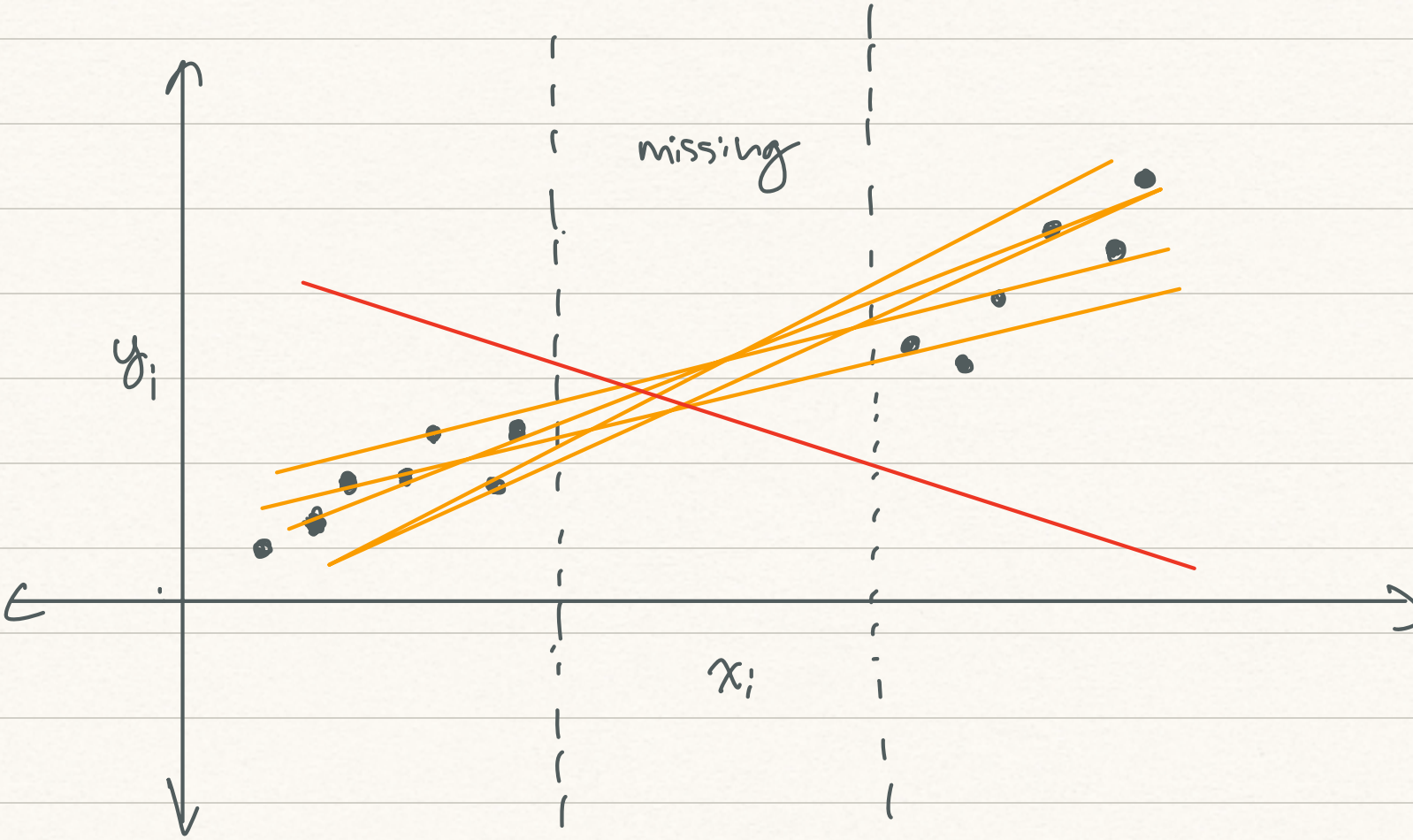
Motivating example

Bayesian linear regression

We have a data set of (x, y) pairs with **censoring** for a certain range of x. (In other words, we never get to observed any (x,y) pair for x in the censored region.)



This might happen if x = time and y = temperature and our thermometer stops recording for some period due to power outage.

This might also happen if x = biomarker, and y = health status, and the only patients that come to the hospital are those with extreme values of the biomarker.

We want to **predict** the relationship between x and y in the censored region.
We cannot do this without **making assumptions** = **modeling**.
There is a clear upward trend in the plot, and let's also say that we have expert background knowledge about (x,y) which supports a monotonic upward trend.

To start we could fit a regression line. However, there are multiple linear trends that all seem plausible and consistent with the data we observe. The orange lines are all consistent with our data, while the red line is not.

In making predictions about the censored region, we would ideally **characterize our uncertainty** about which the true trend. A natural way to accomplish this is by averaging our prediction according to the **posterior probability of each trend**:

Posterior predictive distribution:

$$p(y_{n+1} \mid x_{n+1}, X_{1:n}, Y_{1:n}, -) = \mathbb{E}_{\beta \mid X_{1:n}, Y_{1:n}, -}\left[p(y_{n+1} \mid x_{n+1}, \beta, -)\right]$$

$$= \int_{-\infty}^{\infty} \underbrace{p(\beta \mid X_{1:n}, Y_{1:n}, -)}_{\text{Posterior}} \underbrace{p(y_{n+1} \mid x_{n+1}, \beta, -)}_{\text{Likelihood of new data point}} d\beta$$

This is the motivation for a Bayesian approach to regression. We want to specify our assumptions via a **model** which then gives us a principled way to encode our uncertainty via the **posterior** and **posterior predictive** distributions.

Bayesian linear regression

$$y_i = x_i^T \beta + \varepsilon_i \quad \text{where} \quad \varepsilon_i \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$$

Vectorized form: $\underset{n\times 1}{y} = \underset{n\times p}{X} \underset{p\times 1}{\beta} + \underset{n\times 1}{\varepsilon}$

$$y \equiv \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n, \quad X \equiv \begin{bmatrix} x_1^T \\ \vdots \\ x_n^T \end{bmatrix} \in \mathbb{R}^{n\times p}$$

$$\varepsilon \equiv \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{bmatrix} \in \mathbb{R}^n \quad \text{where} \quad \varepsilon \sim \mathcal{N}(0, \underset{n\times n}{\Sigma}) \quad \Sigma = I\sigma^2$$

multivariate normal

Equivalently: $y \sim \mathcal{N}(X\beta, \Sigma)$

This is an example of Gaussian-Gaussian conjugacy.

Multivariate Gaussian PDF

$$\mathcal{N}(\underset{d\times 1}{y} \mid \underset{d\times 1}{\mu}, \underset{d\times d}{\Sigma}) = \frac{1}{\sqrt{(2\pi)^d \det(\Sigma)}} \exp\left(-\frac{1}{2}(y-\mu)^T \Sigma^{-1} (y-\mu)\right)$$

$$\propto_y \exp\left(-\frac{1}{2} y^T \Sigma^{-1} y + y^T \Sigma^{-1} \mu\right)$$

Kernel of the multivariate Gaussian

Prior:

$$\underset{p\times 1}{\beta} \sim \mathcal{N}(\underset{p\times 1}{m_0}, \underset{p\times p}{L_0^{-1}})$$

prior mean

prior covariance matrix
= inverse of prior precision matrix

Full model:

$$p(y, \beta \mid X, \underbrace{\sigma^2, m_0, L_0}_{\text{"hyperparameters" (fixed or known)}})$$

Posterior calculation:

"and everything else"

$$p(\beta \mid y_{1:n}, X_{1:n}, -) \propto_\beta \mathcal{N}(\beta \mid m_0, L_0^{-1}) \prod_{i=1}^n \mathcal{N}(y_i \mid x_i^T\beta, \sigma^2)$$

$$\propto_\beta \exp\left(-\frac{1}{2}\beta^T L_0 \beta + \beta^T L_0 m_0\right) \prod_{i=1}^n \exp\left(-\frac{1}{2}(y_i - x_i^T\beta)^T \Sigma^{-1} (y_i - x_i^T\beta)\right)$$

$$\propto_\beta \exp\left(-\frac{1}{2}\beta^T L_0 \beta + \beta^T L_0 m_0\right) \prod_{i=1}^n \exp\left(-\frac{1}{2}\beta^T x_i \Sigma^{-1} x_i^T \beta + \beta^T x_i \Sigma^{-1} y_i\right)$$

$$\propto_\beta \exp\left(-\frac{1}{2}\beta^T \underbrace{\left[L_0 + \sum_{i=1}^n x_i \Sigma^{-1} x_i^T\right]}_{\overset{\text{def}}{=} L_n} \beta + \beta^T \underbrace{\left[L_0 m_0 + \sum_{i=1}^n x_i \Sigma^{-1} y_i\right]}_{\overset{\text{def}}{=} L_n m_n}\right)$$

$$\propto_\beta \mathcal{N}(\beta \mid m_n, L_n^{-1})$$

Kernel of a multivariate Gaussian!

posterior precision matrix $L_n = L_0 + \sum_{i=1}^{n} x_i \Sigma^{-1} x_i^T$

posterior mean $m_n = L_n^{-1} L_n m_n$

$$= \left[L_0 + \sum_{i=1}^{n} x_i \Sigma^{-1} x_i^T\right]^{-1} \left[L_0 m_0 + \sum_{i=1}^{n} x_i \Sigma^{-1} y_i\right]$$

Since $\Sigma = I\sigma^2$: $m_n = \left[L_0 + \frac{1}{\sigma^2} X^T X\right]^{-1} \left[L_0 m_0 + \frac{1}{\sigma^2} X^T y\right]$
(in this case)

The MAP (maximum a posteriori) solution is:

$$\hat{\beta}^{MAP} = \underset{\beta}{\operatorname{argmax}}\ \mathcal{N}(\beta \mid m_n, L_n^{-1}) = m_n$$

As the likelihood variance goes to infinity, the MAP becomes the prior mean:

$$\lim_{\sigma^2 \to \infty} \hat{\beta}^{MAP} = L_0^{-1} L_0 m_0 = m_0$$

For simplicity, say that the prior covariance is diagonal and constant:

$$L_0^{-1} = I_p \nu_0^2$$

As the variance of the prior goes to infinity, the MAP becomes the MLE:

$$\lim_{\nu_0 \to \infty} \hat{\beta}^{MAP} = \left[\frac{1}{\sigma^2} X^T X\right]^{-1} \left[\frac{1}{\sigma^2} X^T y\right] = \left[X^T X\right]^{-1} X^T y = \hat{\beta}^{MLE}$$

A Gaussian prior with (near) infinite variance is a "**flat**" or "**non-informative**" prior

Say the prior mean and constant prior variance are equal to the following:

$$L_0^{-1} = I_p \frac{\sigma^2}{\alpha}, \quad m_0 = 0$$

The MAP solution generalizes the **ridge regression** solution:

$$\hat{\beta}^{MAP} = \left[\frac{\alpha}{\sigma^2} + \frac{1}{\sigma^2} X^T X\right]^{-1} \left[\frac{1}{\sigma^2} + X^T y\right] = \left[\alpha + X^T X\right]^{-1} X^T y = \hat{\beta}_{ridge}$$

Why did this happen? **Priors as regularizers:**

$$\hat{\beta}^{MAP} = \underset{\beta}{\operatorname{argmax}}\ p(\beta \mid y, X) = \underset{\beta}{\operatorname{argmax}}\ p(\beta, y \mid X)$$

$$= \underset{\beta}{\operatorname{argmin}}\ -\log p(y \mid X, \beta) - \log p(\beta)$$

$$= \underset{\beta}{\operatorname{argmin}}\ -\sum_{i=1}^{n} \log \mathcal{N}(y_i \mid x_i^T \beta, \sigma^2) - \sum_{j=1}^{p} \mathcal{N}(\beta_j \mid 0, \tfrac{\alpha}{\sigma^2})$$

$$= \underset{\beta}{\operatorname{argmin}}\ \cancel{\frac{1}{2\sigma^2}} \sum_{i=1}^{n} (y_i - x_i^T \beta)^2 + \cancel{\frac{\alpha}{2\sigma^2}} \sum_{j=1}^{p} \beta_j^2$$

(this is the **loss function** for ridge regression with regularization strength alpha)

To summarize: the negative log posterior under a Gaussian-Gaussian model has the same minimizer as the ridge regression loss (i.e. L2-regularized least squares).

This is not an "accident". Priors act as regularizers. The L2 regularization term comes from the Gaussian prior, and a different prior would lead to a different regularizers (e.g., L1 regularization corresponds to a Laplace prior).

Posterior predictive calculation:

$$p(y_{n+1} \mid x_{n+1}, X_{1:n}, y_{1:n}) = \mathbb{E}_{\beta \mid X_{1:n}, y_{1:n}} \left[p(y_{n+1} \mid x_{n+1}, \beta) \right]$$

$$= \int \underbrace{\mathcal{N}(\beta \mid m_n, L_n^{-1})}_{\text{Posterior}} \underbrace{\mathcal{N}(y_{n+1} \mid x_{n+1}^T \beta, \sigma^2)}_{\text{Likelihood of new data point}} d\beta$$

A fact about Gaussians (which we won't prove) is the following:

if $y = Az + b$ and $z \sim \mathcal{N}(\cdots) \rightarrow y \sim \mathcal{N}(\cdots)$

for A, b fixed and known

Applying this to our setting, we have that:

$$y_{n+1} = x_{n+1}^T \beta + \varepsilon_{n+1} \quad \text{where } \beta \sim \mathcal{N}(m_n, L_n^{-1})$$

$$\hookrightarrow \quad y_{n+1} \sim \mathcal{N}(\mu_{n+1}, s_{n+1}^2)$$

Since we know y is Gaussian, we only need to solve for its mean and variance in order to fully describe its distribution.

$$\mu_{n+1} = \mathbb{E}_{\beta,\varepsilon}\left[x_{n+1}^T \beta + \varepsilon_{n+1}\right]$$

$$= x_{n+1}^T \mathbb{E}_{\beta}[\beta] + \mathbb{E}_{\varepsilon_{n+1}}[\varepsilon_{n+1}]$$

$$= x_{n+1}^T m_n$$

$$s_{n+1}^2 = \mathrm{Var}_{\beta,\varepsilon}\left[x_{n+1}^T \beta + \varepsilon_{n+1}\right]$$

$$= x_{n+1} \mathrm{Cov}(\beta) x_{n+1}^T + \mathrm{var}(\varepsilon_{n+1})$$

$$= x_{n+1} L_n^{-1} x_{n+1}^T + \sigma^2$$

<u>Marginal likelihood and prior predictive distribution:</u>

What if there are no training data points? $(n=0)$

$$\mu_{n+1} = x_{n+1}^T m_n = x_{n+1}^T [L_0]^{-1} [L_0 m_0] = x_{n+1}^T m_0$$

$$s_{n+1}^2 = x_{n+1} L_n^{-1} x_{n+1}^T = x_{n+1} L_0^{-1} x_{n+1}^T$$

The "posterior" parameters are just the **prior parameters** when n=0

The posterior predictive becomes the "prior predictive" when n=0:

$$P(y_{n+1} \mid x_{n+1}, y_{1:n}, x_{1:n})$$

$$= P(y_1 \mid x_1)$$ ...which is just the marginal likelihood of a single data point

...which is Gaussian:

$$= \mathcal{N}(y_1 \mid x_1^T m_0, x_1 L_0 x_1^T)$$

The marginal likelihood as a sequence of posterior predictives:

$$p(y_{1:n} \mid X_{1:n}) = \prod_{i=1}^{n} p(y_i \mid x_i, y_{<i}, X_{<i})$$

$$= \prod_{i=1}^{n} \mathcal{N}(y_i \mid x_i^T m_{i-1}, x_i L_{i-1}^{-1} x_i^T)$$

$$L_{i-1} = L_0 + \frac{1}{\sigma^2} \sum_{k<i} x_k x_k^T$$

$$m_{i-1} = L_i^{-1} [L_0 m_0 + \frac{1}{\sigma^2} \sum_{k<i} x_i y_i]$$

Bayesian updating: for each data point, the new "prior" is the posterior given all the data up to that point.

An alternative way to derive the marginal likelihood uses the vectorized form of the likelihood for all n data points:

$$\underset{n\times 1}{y} = \underset{n\times p}{X} \underset{p\times 1}{\beta} + \underset{n\times 1}{\varepsilon} \quad \text{where} \quad \underset{p\times 1}{\beta} \sim \mathcal{N}(\underset{p\times 1}{m_0}, \underset{p\times p}{L_0^{-1}}) \rightarrow y \sim \mathcal{N}(\underset{n\times 1}{\mu}, \underset{n\times n}{S})$$

$$\mu = \mathbb{E}[X\beta + \varepsilon] = X m_0$$

$$S = \mathrm{cov}(X\beta + \varepsilon) = X \,\mathrm{cov}(\beta) X^T + \mathrm{cov}(\varepsilon) = X L_0^{-1} X^T + I_n \sigma^2$$

So the **marginal likelihood** can be expressed as a product of univariate Gaussians (each of which is a posterior predictive) or as a single multivariate Gaussian.

Model evaluation via the prior/posterior predictive distribution

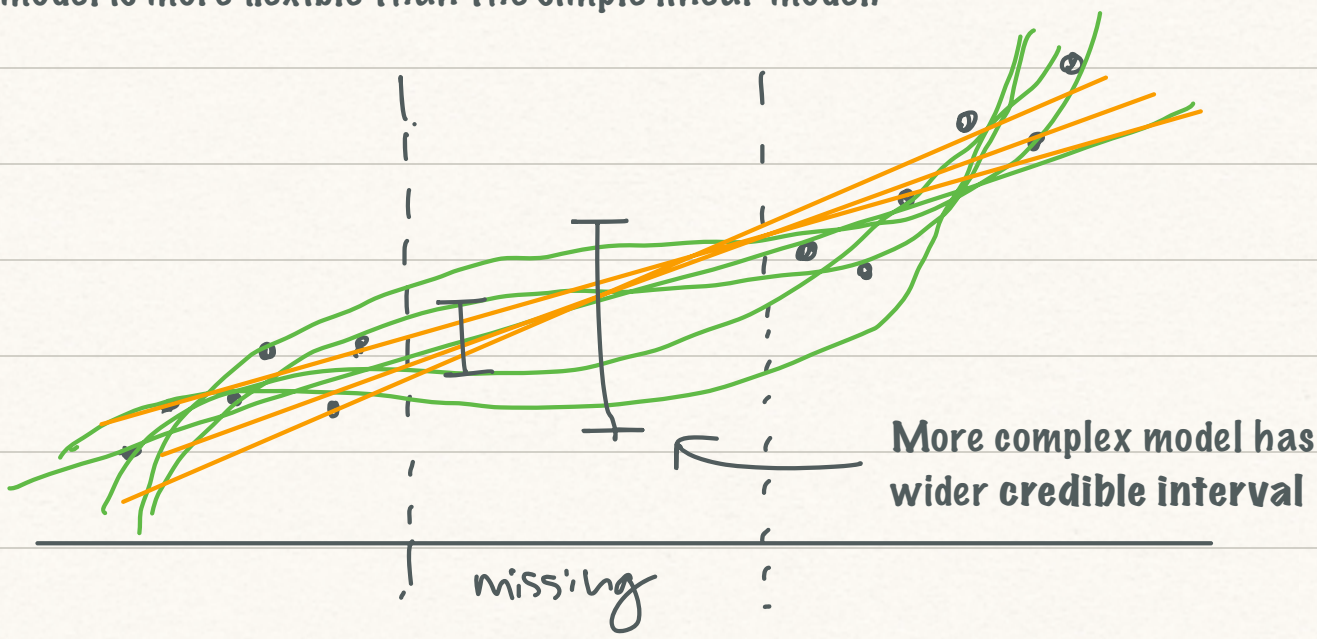
Consider a more complex model class for Bayesian linear regression:

$$P(y \mid x, \beta) = \mathcal{N}(y;\ \text{poly}_3(x)^T \beta,\ \sigma^2)$$

e.g. if $x \in \mathbb{R} \rightarrow \text{poly}_3(x) = [x, x^2, x^3]$

$\hookrightarrow \text{poly}_3(x)^T \beta = \beta_1 x + \beta_2 x^2 + \beta_3 x^3$

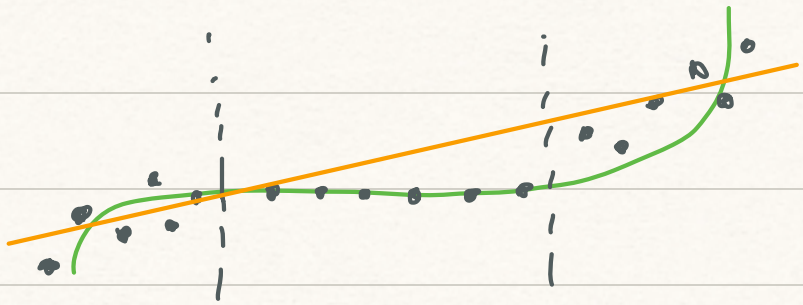
This model is more flexible than the simple linear model:



All of the green curves correspond to a possible values of beta in the cubic (more complex) model which give high-ish likelihood to the observed data points.

All of the orange curves correspond to possible values of beta in the linear (simple) model which give high-ish likelihood to the observed data points.

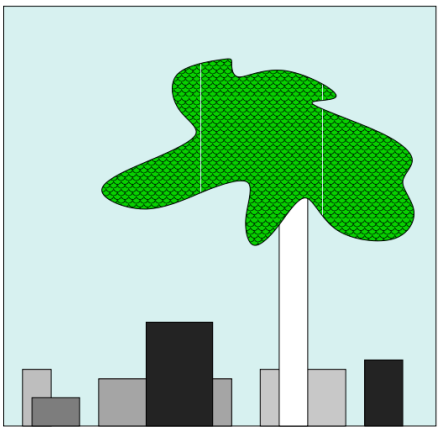
Certain models (i.e., curves) in the complex model class **could** fit the heldout data points much better than any plausible model in the linear class...



...however there are **many more** models in the complex model class that are consistent with the training data.

The principle of "Occam's razor" says we should take the simplest hypothesis that is consistent with the data. In this case a model class is a hypothesis and the model evidence (aka the marginal likelihood) accounts for simplicity.

See Chapter 28 of Mackay:



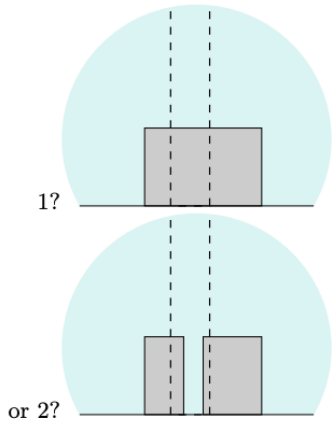
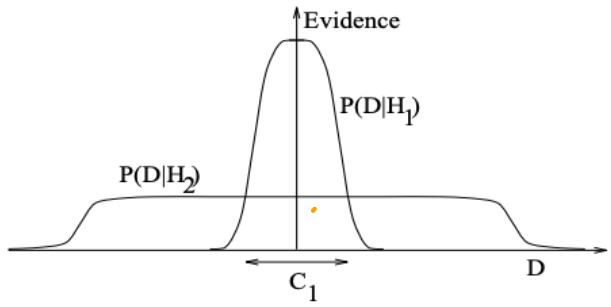


Figure 28.2. How many boxes are behind the tree?



Probabilistic models, in general:

$$P(D, Z \mid H)$$

modeling assumptions (including hyperparameters)

data

latents

likelihood

prior

Posterior

$$P(Z \mid D, H) = \frac{P(D \mid Z, H)\, P(Z \mid H)}{P(D \mid H)}$$

marginal likelihood
model evidence
normalizing constant
prior predictive distribution
...

When comparing two model classes (H1 vs H2), the **Bayes factor**, rewards simplicity:

$$BF = \frac{P(D \mid H_1)}{P(D \mid H_2)}$$

Similarly, we can compare **posterior** (rather than prior) **predictive** probabilities, if we create a train-test split of the data:

$$P(D^{test} \mid D^{train}, H_1) \quad \text{vs.} \quad P(D^{test} \mid D^{train}, H_2)$$

$$P(D^{test} \mid D^{train}, H) = \int P(D^{test} \mid D^{train}, z, H)\, P(z \mid D^{train}, H)\, dz$$

Model selection via the prior/posterior predictive distribution:

Model evaluation and selection are naturally related. Just as it makes sense to use the marginal likelihood to evaluate, it also makes sense to use it to select.

A common example of this are **empirical Bayesian** procedures for choosing the prior in a data-driven manner.

For example, for the Bayesian regression model above, the marginal likelihood is:

$$p(Y_{1:n} \mid X_{1:n}, \sigma^2, \underbrace{m_0, L_0})$$

hyperparameter for the likelihood

hyperparameters for the prior

For example, for fixed values of the other hyperparameters we could do **type-II maximum likelihood** to set the prior mean:

$$m_0 \leftarrow \underset{m_0}{\operatorname{argmax}}\; p(Y_{1:n} \mid X_{1:n}, \sigma^2, m_0, L_0)$$